AHP–TOPSIS Multi-Criteria Decision Visibility Engine in Agrivive

I. INTRODUCTION

Agrivive is a surplus-vegetable marketplace. Its visibility engine (*visibility-ahptopsis-v1*) decides which listings buyers see first and which listing is boosted to Facebook and Instagram. It combines two multi-criteria decision-making methods. The Analytic Hierarchy Process (AHP) turns pairwise expert judgments into criterion weights and checks them with a consistency ratio below 0.10 [1]. TOPSIS then ranks each listing by its relative closeness to an ideal best listing and its distance from an ideal worst listing [2]. The hybrid has been applied to agricultural decisions [3], and TOPSIS has been reviewed for agricultural use [4]. Pairwise-weight consistency is also an active concern in the AHP literature [5].

The problem is that produce is perishable and supply is uneven. Newest-first, cheapest-first or popularity-first sorting hides aging, discounted or large-stock listings and favors big sellers. Online ranking research shows that how products are ordered shapes outcomes on a platform [6]. The engine scores every eligible listing on four raw criteria and orders the feed by the result. This gives Agrivive a repeatable, explainable ranking that pushes exposure toward produce that most needs to sell. The design is described in the project's capstone manuscript [7].

II. RULES

Criteria. All four are raw values, and the engine treats every one as "bigger is better."

#	Criterion	Raw input
C1	Posting age	Hours since the listing was published
C2	Price drop	Base price − current price (pesos, floored at 0)
C3	Stock level	Quantity left
C4	Seller reliability	Average review rating, 1–5; 2.5 if no reviews

Eligibility.

- The listing is in its latest cycle, active and marketable.
- Quantity and price are above 0.
- The seller is active and email-verified.
- The seller has a complete public profile: shop name, address, phone of at least 7 characters, and valid coordinates.

Safety rules.

- Unusable inputs (NaN, negative) become 0.
- If CR > 0.10 the engine throws and refuses to rank.
- Equal scores break alphabetically by id.
- A lone or all-identical set of listings scores 0.5 each.
- Policy switch: *VISIBILITY_POLICY=visibility-v4* rolls back to the old formula; any other value uses AHP-TOPSIS.

Boost rule (six-hour lock). Go down the ranking and pick the first listing whose last boost was at least 6 hours ago. Locked listings are skipped even if they rank first. The backend decides, and n8n only posts.

III. WEIGHTS (AHP)

The Saaty-scale pairwise matrix **A** is:

	C1	C2	C3	C4
C1	1	1	2	3
C2	1	1	2	3
C3	1/2	1/2	1	2
C4	1/3	1/3	1/2	1

Column sums are 2.833, 2.833, 5.5 and 9. Normalizing each cell and averaging each row gives:

Criterion	Weight
Posting age	0.3507 (35.1%)
Price drop	0.3507 (35.1%)
Stock level	0.1892 (18.9%)
Seller reliability	0.1093 (10.9%)

The weights sum to 1. The consistency ratio is **CR = 0.0038 < 0.10**, so the judgments are accepted [1]. Age and price drop each count roughly twice as much as stock and three times as much as seller reliability.

IV. FORMULAS

AHP weights, with n = 4:

$$w_i = (1/n)\, \Sigma_j\, a_{ij} \,/\, \Sigma_k\, a_{kj} \tag{1}$$

$$\lambda_{max} = (1/n)\, \Sigma_i\, (\boldsymbol{Aw})_i \,/\, w_i, \;\; CI = (\lambda_{max} - n)/(n-1), \;\; CR = CI \,/\, RI, \;\; RI = 0.90 \tag{2}$$

TOPSIS [2], for listing i and criterion j:

$$r_{ij} = x_{ij} \,/\, \sqrt{(\Sigma_i\, x_{ij}^2)} \tag{3}$$

$$v_{ij} = w_j \cdot r_{ij} \tag{4}$$

$$A_j^{+} = max_i\, v_{ij}, \quad A_j^{-} = min_i\, v_{ij} \tag{5}$$

$$D_i^{+} = \sqrt{(\Sigma_j\, (v_{ij} - A_j^{+})^2)}, \quad D_i^{-} = \sqrt{(\Sigma_j\, (v_{ij} - A_j^{-})^2)} \tag{6}$$

$$C_i = D_i^{-} \,/\, (D_i^{+} + D_i^{-}), \quad 0 \le C_i \le 1 \tag{7}$$

The seller card shows the score as $100 \times C_i$, to one decimal place.

V. PROCESS

1. **Collect.** Load all eligible listings and compute the four raw criteria for each.
2. **Weigh.** Compute AHP weights from the pairwise matrix and check CR (Eqs. 1–2). Stop if CR > 0.10.

3. **Build the matrix.** One row per listing and one column per criterion.
4. **Normalize.** Divide each column by its length (Eq. 3). An all-zero column stays 0.
5. **Weight.** Multiply each column by its AHP weight (Eq. 4).
6. **Find the ideals.** Take the best and worst value in each column (Eq. 5).
7. **Measure.** Compute each listing's distance to the best and to the worst (Eq. 6).
8. **Score and sort.** Compute closeness (Eq. 7) and sort descending, with ties broken by id.
9. **Apply.** The feed uses the sorted order. For a boost, the highest-ranked listing not boosted in the last 6 hours is chosen.

VI. EXAMPLE SCENARIO AND COMPUTATION

Four listings in the same feed:

Listing	Hours posted	Start price	Price now	Drop	Stock	Rating
A	12	₱40	₱32	8	40	4.8
B	8	₱30	₱26	4	25	4.5
C	4	₱25	₱25	0	60	4.0
D	1	₱20	₱20	0	15	3.6

Step 1: column lengths (denominator of Eq. 3). C1 = $\sqrt{(12^2 + 8^2 + 4^2 + 1^2)} = \sqrt{225} = 15.0000$. C2 = $\sqrt{(8^2 + 4^2)} = \sqrt{80} = 8.9443$. C3 = $\sqrt{(40^2 + 25^2 + 60^2 + 15^2)} = \sqrt{6050} = 77.7817$. C4 = $\sqrt{(4.8^2 + 4.5^2 + 4.0^2 + 3.6^2)} = \sqrt{72.25} = 8.5000$.

Step 2: normalized values r_{ij}. Example: A's C1 = 12 / 15 = 0.8000.

	C1	C2	C3	C4
A	0.8000	0.8944	0.5143	0.5647
B	0.5333	0.4472	0.3214	0.5294
C	0.2667	0.0000	0.7714	0.4706
D	0.0667	0.0000	0.1928	0.4235

Step 3: weighted values v_{ij}. Example: A's C1 = 0.8000 × 0.3507 = 0.2806.

	C1	C2	C3	C4
A	0.2806	0.3137	0.0973	0.0617
B	0.1870	0.1568	0.0608	0.0579
C	0.0935	0.0000	0.1460	0.0514
D	0.0234	0.0000	0.0365	0.0463

Step 4: ideal solutions.

	C1	C2	C3	C4
A^+ (best)	0.2806	0.3137	0.1460	0.0617
A^- (worst)	0.0234	0.0000	0.0365	0.0463

Step 5: distances for A (Eq. 6). $D_A^+ = \sqrt{(0^2 + 0^2 + (0.1460 - 0.0973)^2 + 0^2)} = 0.0487$. $D_A^- = \sqrt{(0.2572^2 + 0.3137^2 + 0.0608^2 + 0.0154^2)} = \sqrt{0.1685} = 0.4105$.

Step 6: closeness for A (Eq. 7). $C_A = 0.4105 / (0.0487 + 0.4105) =$ **0.894**. Repeating the steps for the other listings gives the ranking:

Rank	Listing	D^+	D^-	Closeness	Score shown
1	A	0.0487	0.4105	0.8940	89.4
2	B	0.2015	0.2283	0.5311	53.1
3	C	0.3654	0.1301	0.2626	26.3
4	D	0.4204	0.0000	0.0000	0.0

Interpretation. A is the oldest, has the largest price drop and the best rating, so it is closest to ideal and ranks first. D is newest, has no drop, the least stock and the lowest rating, so it is the worst listing in the set and scores 0.

Sensitivity.

- Changing A's age from 12 hours to 1 hour drops A to 53.7 and lifts B to 63.8, so B takes rank 1.
- Giving D an ₱8 price drop lifts D to 45.5 and moves it from rank 4 to rank 3.
- Cutting B's stock from 25 to 2 moves B only about 4 points (53.1 to 49.1), because stock carries 18.9% of the weight.

Edge cases the engine passes.

Case	Score
Best in every column	100.0
Worst in every column	0.0
Only one listing	50.0
Four identical listings	50.0 each
All values zero	50.0 each, no crash

Boost application. If A was boosted 3 hours ago, it is locked until 6 hours have passed. The boost selector skips it and chooses B, the next-ranked listing that is not locked.

VII. CONCLUSION

The engine gives Agrivive a transparent, defensible visibility policy. Expert judgment becomes weights that pass a consistency check (CR = 0.0038) [1]. TOPSIS then ranks the live feed by closeness to the ideal listing [2]. Surplus, discounted and well-reviewed produce rises, and the six-hour lock keeps one listing from being boosted repeatedly. The method rests on established decision theory with agricultural precedent [3], [4]. The weights are still prototype judgments that can be recalibrated with local stakeholder surveys.

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